

ARTIFICIAL INTELLIGENCE

ENCT 351

Year/Part: III/II

Teaching Schedule				Examination Scheme						Total I
L	T	P	Total	Theory			Practical			
				Assessment Marks	Final		Assessment Marks	Final		
					Duration (Hrs)	Marks		Duration (Hrs)	Marks	
3	1	3	7	40	3	60	25	0	0	125

Depth Codes

E-Explanation	C-Circuit	D-Definition	DM-Demonstration
DV-Derivation	DW-Drawing	P-Proof	I-Illustration
NUM-Numerical	PRG-Programming	S-State	ACT-Activity-based Learning
MP- Mini Project	EXP-Experiment	REV-Review / Recap	PS- Problem Solving
QA- Question Answer	Q- Quiz	ST- Surprise Test	MT-Mid Term Test

Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan			Week
				L	T	P	
1	Introduction			4	1	3	1,2
	1.1. Definition, Foundation, History of AI	D, E	Definition of AI, Turing test, Reverse Turing Test, Total Turing test Foundation & History of AI	0.5			
	1.2. Definition and importance of Knowledge	D, I	Definition, importance and examples of Knowledge DIKW Pyramid Data, Information, Knowledge, Intelligence	0.5			
	1.3 Cognition and Learning	D, E	Neuroscience, Human Cognition, Learning, Cognitive computing				
	1.4. Human intelligence and Machine intelligence	D, E	State of the Art What Humans and AI can do? Comparison based on various parameters	0.5			
	1.5. AI Tree: Branches and Interdisciplinary Nature	D, E	AI as a tree with branches and core concepts Applications in interdisciplinary areas	0.5			
	1.6. Intelligent Agents and Types	D, E, PS	Definition of Intelligent Agent, Block Diagram, explanation, factor considerations while selections an agent, exercises about different type of agents	2	1	3	
	1.7 Agentic AI*		Definition, characteristics of Agentic AI and Autonomous Intelligence				
	Evaluation	QA, Q					
Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan			Week
				L	T	P	
2	Problem Solving and Search			9	3	=	2,3,4,5
	2.1. Formal problem definition: States,	D, E, NUM	Definition of Problem-solving agent, States, State Space and Problem Space	1	1		

	actions, transitions, well-defined problems		Problem Representation using Graph vs Tree Well-defined problems and solutions Example Problems				
	2.2. Constraint Satisfaction Problems: Node Consistency, Path Consistency, Backtracking	E, I, NUM	Defining CSPs; Constraint Propagation; Node Consistency, Path Consistency, Arch Consistency, Backtracking Search with examples (Fixed Examples topics: Cryptarithmic, Map coloring, Eight Queens or mini-Sudoku) Variable Limitation--□ Easy/Medium	1			
	2.3. Search algorithms, Strategies, and Evaluations	D, E,	Search algorithms, Strategies, and Evaluations *relation to neural networks (eg: alpha go) *beam search	0.5		3	
	2.4. Uninformed: BFS, DFS, Iterative Deepening	D, E, NUM	Definitions and Complexity of BFS, DFS, Iterative Deepening (review)	1 /0.5			
	2.5. Informed Search: Best First search, Greedy Search, A* algorithm	D, E, NUM	Definitions and Examples : Best First search, Greedy Search, A* algorithm	1	1		
	2.6. Adversarial search: Minimax algorithm, Alpha-beta pruning	D, E, NUM	Definitions, Algorithms and Example of Adversarial search: Minimax algorithm, Alpha-beta pruning	1			
	2.7. Local Search and Optimization: Hill climbing, Simulated Annealing	D, E, I	Definitions , Algorithms and Example of Hill climbing, Simulated Annealing	1.5		3	
	2.8. Evolutionary Optimization: Genetic Algorithm	D, E, NUM	Definitions , Algorithms and Example of Genetic Algorithm	1.5	1		
	Evaluation	QA, Q					
Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan			Week
				L	T	P	
3	Knowledge Representation and Probabilistic Reasoning			7	2	6	5,6,7
	3.1. Knowledge-based Agent	D, E, I	Definition of Knowledge-Based Agents; Example: The Wumpus World;	0.5			
	3.2. Knowledge Representation Techniques and Issues in Representation	D, E, I	Knowledge Engineering Approaches for Knowledge Representation Issues in Representation Path planning	1		3	
	3.3. Propositional & Predicate Logic	REV, E, NUM	Review of Propositional Logic Rules of Inference, Predicate Logic, and Resolution Refutation Forward and Backward Chaining	1.5	2		
	3.4. Semantic networks and frames	D, E, NUM	GUI for Logic Definition of and Examples of Semantic networks and frames	0.5			

			*Knowledge graph				
	3.5. Probabilistic Reasoning, Inference using Bayes' Theorem	E, DV, NUM	***Review of Bayes Network and Probabilistic Reasoning (Numerical)	1			
	3.6. Fuzzy logic: membership functions, fuzzy inference systems	D, E, I	Concept of Fuzzy Logic, Fuzzy Sets vs. Crisp Sets, Role of membership functions and Common Shapes Components of a Fuzzy Inference System (FIS). Applications	2			
	Evaluation	QA, Q					
Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan			Week
				L	T	P	
4	Machine Learning Fundamentals			10	5	6	7,8,9, 10
	4.1. Foundations and Four pillars of Machine Learning	D, E, I	From Book Deisenroth, Faisal & Ong, <i>Mathematics for Machine Learning</i>, Cambridge University Press	1			
	4.2. Review of Mathematics for Machine Learning	E, I, NUM	Linear Algebra for Machine learning (Vector Space, Linear Combinations, Basis)	2.5	1		
	4.3 Continuation Optimization; Unconstraint Optimization, Constraint Optimization, Convex Optimization	E, I	Unconstraint Optimization, Review of Stochastic gradient decent Algorithm Beyond gradient decent Jacobians, Hessians, Calculating derivatives of vector valued function Constraint Optimization, Convex Optimization	0.5			
	4.2.Review of Machine Learning Pipeline		4.4.1. Model Development: Generative vs. Discriminative: 4.4.2. Learning Algorithm 4.4.3. Capacity, Overfitting and Underfitting 4.4.4. Hyperparameters and Validation Sets 4.4.5. Estimators, Bias and Variance 4.4.6. Review of MLE & MAP and Cross Entropy				
	4.5 Supervised Learning Algorithm: Decision Tree and Support Vector Machine	E, DV, NUM,MP, PRG	Support Vector Machine, Decision Trees	3	1		
	4.6 Unsupervised Learning Algorithm: t-SNE	E, NUM,MP, PRG	t-SNE	2			
	4.7 Semi Supervised and Reinforcement Learning	E, I, NUM	Introduction and Examples of Semi supervised learning Reinforcement Learning (Q-Learning)	1	2		

	4.8 Model Evaluation: Energy-based Indicators		- Training time or energy - Model size - Inference time or energy				
	Evaluation	QA, Q					
Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan			Week
				L	T	P	
5	Neural Networks and Deep Learning Algorithms			6	2	3	11,12
	5.1. Neural Networks: Structures and Activation Functions	D, E	Biological vs. Artificial Neurons: Analogy and differences. Right Brain/Left Brain A simple mathematical model for a neuron. Neural Networks: Structures Why Learning Converges: Universal Approximation Theorem Learning in NN. (Hebbian, Logic gates) Types of Activation Functions its purpose and Characteristics	1.5	1		
	5.2. Perceptron, Multilayer Perceptron, and Backpropagation	D, E, DV	Perceptron and its limitations. Single-layer feed-forward neural networks Architecture and function of a Multilayer Perceptron (MLP). The Backpropagation algorithm and it's derivation	2	1		
	5.3. Introduction to Deep learning	D, E	ML Vs DL Deep Learning and its distinction from traditional Neural Networks. Building Blocks of Deep Neural Networks Forward and Backward Propagation Why Deep Representations? Vectorized Implementation. Getting your Matrix Dimensions Right Prerequisites for successful Deep Learning. https://medium.com/@tharangachaminda/neural-network-06-deep-l-layer-neural-network-d843c20b0f3c Introduction to CNN, Components Applications*	1.5			
	5.4. Concepts on Recurrent and Generative Neural Networks	D, E	Introduction to RNN, Components Applications (Recurrent Neurons and Recurrent Layer) * History of Generative Models Markov Models Introduction to GANs, Components Applications*	1			
	Evaluation	QA, Q					
Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan			Week
				L	T	P	
6	AI Applications			7	2	3	13,14, 15

	6.1. Expert Systems: Characteristics, Architecture, Development and Various Applications	D, E	Definition, characteristics, Architecture and components of Expert Systems Pros and Cons Development Phases. Applications	1			
	6.2. NLP: Level of Analysis and Application	D, E, MP, PRG	Levels of analysis in NLP with examples Phonological/Morphological Analysis. Lexical Analysis, Syntactic Analysis, Semantic Analysis and Pragmatic Analysis, Discourse Analysis Challenges in NLP applications Embeddings & Modern NLP models.*	2.5	1		
	6.3. Robotics & Computer Vision: Fundamental, Components and Applications	D, I, E, MP, PRG	Robotics and its fundamental components. Robot Classifications Computer Graphics/Image Processing Computer Vision Computer Vision (CV) and its core concepts. Image Filtering/Enhancement, Feature Extraction (Edges), Object Detection and Recognition. Discuss the applications of Robotics and Computer Vision. Robotics: Industrial automation, surgical assistance, space exploration. CV: Autonomous vehicles, facial recognition, medical imaging analysis, quality control. Use of Deep Learning *	2.5	1		
	6.4. Sustainable AI systems	D, E	Concept of sustainable AI systems. Using AI to optimize energy grids, predict climate change, and improve resource efficiency. Energy Consumption of training large models and potential for "Green AI" practices.	1			
	Evaluation	QA, Q					
Unit	Topic/ Sub topic	Depth Code	Description of Depth	Actual plan			Week
				L	T	P	
7	Emerging Trends			2			15
	Sequence to Sequence Models	D, E	Concept, benefits Application of Sequence to Sequence Models	0.5			
	Federated Learning	D, E	Concept, benefits Application of Federated Learning	0.5			
	Edge AI	D, E	Concept, benefits Application of Edge AI	0.5			
	Ethics and AI: Responsible AI	D, E, ACT	Definition of Responsible AI and its core pillars. Framework for developing and deploying AI Concepts of bias and fairness in AI. Introduction to Explainable AI (XAI). Privacy risks associated with AI Introduction to AI Security threats e.g.: Adversarial Attacks, Poisoning e.t.c and Defense Strategies	0.5			

			Activity includes: Policy/Regulatory Framework Review and Presentation				
	Evaluation	QA, Q					

*Descriptive Notes Questions Only

Tutorial

The following tutorial activities (including, but not limited to, those listed) should be completed to cover all the required content of this course.

- Design a simple agent model (vacuum cleaner, chess player).
- Formulate a route-finding problem using states, actions, and transitions.
- Solve the Problem with uninformed strategies
- Apply the A* algorithm with the given heuristic values.
- Solve the minimax tree and demonstrate alpha–beta pruning
- Solve a CSP like map coloring or mini-Sudoku.
- Perform one iteration of the Genetic Algorithm (selection, crossover, mutation).
- Represent facts using semantic networks and frames.
- Solving problems related to Bayes' theorem & Fuzzy Logic
- Understanding machine learning basics through model training, cross-validation, and performance evaluation.
- Calculation related to Decision Tree and Support Vector Machine
- Model Evaluation: Build a confusion matrix. Calculate Sensitivity, Specificity, Precision, Recall, and F1-score.
- Solve perceptron weight updates for a small dataset.
- Derive forward and backward propagation numerically for a 2-layer NN.
- Case study: Identify ethical risks in real-world AI use—responsible AI in Nepal's context.

Practical

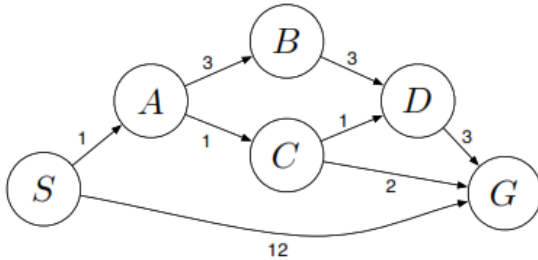
The following lab exercises (including but not limited to those listed) will give students hands-on experience in applying artificial intelligence and machine learning tools to solve real-world problems:

- Week 1: Intelligent Agents & Problem Formulation
- Week 2: Uninformed Search Techniques
- Week 3: Informed (Heuristic) Search Techniques
- Week 4: Adversarial Search & Game Playing
- Week 5: Constraint Satisfaction Problems (CSP)
- Week 6: Evolutionary Computation (Genetic Algorithms)
- Week 7: Knowledge Representation (Logic, Semantic Networks, Frames)
- Week 8: Probabilistic & Fuzzy Reasoning
- Week 9: Machine Learning Pipeline & Data Preprocessing
- Week 10: Supervised & Unsupervised Learning
- Week 11: Neural Networks & Deep Learning Basics
- Week 12: Mini Project, AI Applications & Ethics

References: (Primarily based on the syllabus, and relevant chapters may be consulted as needed)

1. Stuart Russell & Peter Norvig, *Artificial Intelligence: A Modern Approach*, Pearson (4th edition)
2. Elaine Rich, Kevin Knight, Shivashankar B. Nair, *Artificial Intelligence, 3rd Edition*
3. Christopher Bishop, *Pattern Recognition and Machine Learning*, Springer
4. Deisenroth, Faisal & Ong, *Mathematics for Machine Learning*, Cambridge University Press
5. Ian Goodfellow, Yoshua Bengio, Aaron Courville, *Deep Learning*, MIT Press, 2016

Model Question**ARTIFICIAL INTELLIGENCE****ENCT 305****Year/Part: III/I**

QN	Question	Marks	Unit														
1.	What is Agentic AI? How Intelligent agent and Agent AI is contributing to Autonomous Intelligence?	1+4 =5	1														
2.	Why A* is preferred over greedy search? For following graph, compute A* search. Assume start state is S and goal state is G. <table data-bbox="932 692 1121 974"><thead><tr><th>State</th><th>h_1</th></tr></thead><tbody><tr><td>S</td><td>5</td></tr><tr><td>A</td><td>3</td></tr><tr><td>B</td><td>6</td></tr><tr><td>C</td><td>2</td></tr><tr><td>D</td><td>3</td></tr><tr><td>G</td><td>0</td></tr></tbody></table>	State	h_1	S	5	A	3	B	6	C	2	D	3	G	0	2+4=6	2
State	h_1																
S	5																
A	3																
B	6																
C	2																
D	3																
G	0																
3.	A diagnostic AI system uses the following rules: ● If battery voltage is low AND starter is inactive → car won't start ● If fuel level is low AND battery voltage is normal → engine cranks but doesn't start ● If car won't start AND engine cranks slowly → battery problem Facts given: ● Battery voltage is low ● Starter is inactive ● Engine cranks slowly Using forward chaining, prove whether the system concludes a battery problem.	6	2														
4.	How is inference carried out using Bayes' theorem? In a study of pleas and prison sentences, it is found that 45% of the subjects studied were sent to prison. Among those sent to prison, 40% chose to plead guilty. Among those not sent to prison, 55% chose to plead guilty. a) If one of the study subjects is randomly selected, find the probability of getting someone who was not sent to prison. b) If a study subject is randomly selected and it is then found that the subject entered a guilty plea, find the probability that this person was not sent to prison.	2+4=6	3														
5.	When do we use Naive Bayesian classifier? The data samples are described by attributes age, income, student, and credit. The class label attribute, buys, tells whether the person buys a computer, has two distinct values, yes (class C1) and no (class C2). Using the Naive Bayesian classifier, classify sample X = (age = youth, income = medium, student = yes, credit = fair)	2+2+2=6	4														

	<table><tr><th>RID</th><th>age</th><th>income</th><th>student</th><th>credit</th><th>C_i: buy</th></tr><tr><td>1</td><td>youth</td><td>high</td><td>no</td><td>fair</td><td>C_2: no</td></tr><tr><td>2</td><td>youth</td><td>high</td><td>no</td><td>excellent</td><td>C_2: no</td></tr><tr><td>3</td><td>middle-aged</td><td>high</td><td>no</td><td>fair</td><td>C_1: yes</td></tr><tr><td>4</td><td>senior</td><td>medium</td><td>no</td><td>fair</td><td>C_1: yes</td></tr><tr><td>5</td><td>senior</td><td>low</td><td>yes</td><td>fair</td><td>C_1: yes</td></tr><tr><td>6</td><td>senior</td><td>low</td><td>yes</td><td>excellent</td><td>C_2: no</td></tr><tr><td>7</td><td>middle-aged</td><td>low</td><td>yes</td><td>excellent</td><td>C_1: yes</td></tr><tr><td>8</td><td>youth</td><td>medium</td><td>no</td><td>fair</td><td>C_2: no</td></tr><tr><td>9</td><td>youth</td><td>low</td><td>yes</td><td>fair</td><td>C_1: yes</td></tr><tr><td>10</td><td>senior</td><td>medium</td><td>yes</td><td>fair</td><td>C_1: yes</td></tr><tr><td>11</td><td>youth</td><td>medium</td><td>yes</td><td>excellent</td><td>C_1: yes</td></tr><tr><td>12</td><td>middle-aged</td><td>medium</td><td>no</td><td>excellent</td><td>C_1: yes</td></tr><tr><td>13</td><td>middle-aged</td><td>high</td><td>yes</td><td>fair</td><td>C_1: yes</td></tr><tr><td>14</td><td>senior</td><td>medium</td><td>no</td><td>excellent</td><td>C_2: no</td></tr></table>	RID	age	income	student	credit	C_i : buy	1	youth	high	no	fair	C_2 : no	2	youth	high	no	excellent	C_2 : no	3	middle-aged	high	no	fair	C_1 : yes	4	senior	medium	no	fair	C_1 : yes	5	senior	low	yes	fair	C_1 : yes	6	senior	low	yes	excellent	C_2 : no	7	middle-aged	low	yes	excellent	C_1 : yes	8	youth	medium	no	fair	C_2 : no	9	youth	low	yes	fair	C_1 : yes	10	senior	medium	yes	fair	C_1 : yes	11	youth	medium	yes	excellent	C_1 : yes	12	middle-aged	medium	no	excellent	C_1 : yes	13	middle-aged	high	yes	fair	C_1 : yes	14	senior	medium	no	excellent	C_2 : no		
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6.	Describe the machine learning pipeline with an example. Discuss preprocessing, splitting, training, model tuning, and final evaluation. OR Discuss the different approaches for validating a classifier with calculating the accuracy of this Covid case test data. A confusion matrix for covid testing classifier is as follows: <table><tr><th colspan="2" rowspan="2"></th><th colspan="2">Predicted Covid Cases</th></tr><tr><th>True</th><th>False</th></tr><tr><th rowspan="2">Actual Covid Cases</th><th>True</th><td>456</td><td>52</td></tr><tr><th>False</th><td>78</td><td>11569</td></tr></table> Is accuracy sufficient to indicate the performance of this classifier? Justify with calculations and comparison of other parameters like precision, recall and F-1 scores.			Predicted Covid Cases		True	False	Actual Covid Cases	True	456	52	False	78	11569	3+3=6 2+4=6	4																																																																													
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Actual Covid Cases	True	456	52																																																																																										
	False	78	11569																																																																																										
7.	“Deep learning models sometimes outperform traditional computer vision methods” What are the key factors responsible for this? Why Generative Neural Networks is popular? Describe briefly its components	3 4	5																																																																																										
8.	When do we use back propagation algorithm? Perform derivation for backward propagation to minimize the error.	1+5	5																																																																																										
9.	Compare and contrast i. NLU Vs NLG ii. Machine learning vs Federated learning	(3*2) =6	6,7																																																																																										
10.	Write short notes on the following: - iii. Fuzzy Logic iv. Responsible AI	(3*2) =6	3,7																																																																																										